

High-Value AI Use Cases for IAM

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Key Issues



Evaluation of
AI for fit for purpose



Use cases where
AI does not fit



High value
AI use cases

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AI Assistant and Agent Fit for Purpose

Criteria	Good for AI?
Can be done well (enough) by humans: How to do it is generally known	
Priority is acceleration	
Requires clear human accountability	
Priority is quality	
Requires real-time or near-real-time responsiveness	
Requires deterministic decisions	

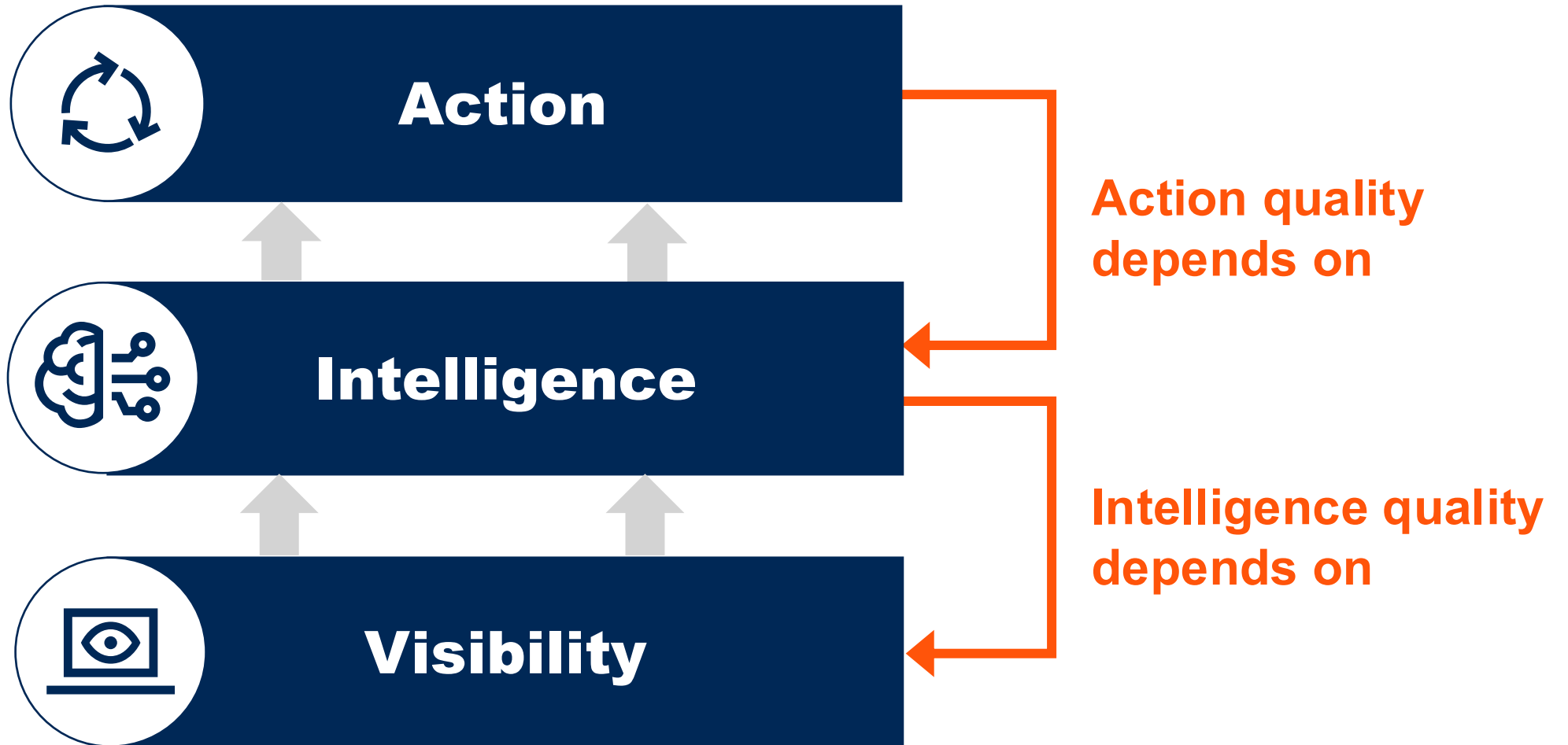


Use Value and Feasibility to Prioritize AI Implementation

- »»» Value — What tangible outcome would applying AI to this process achieve?
- 👍 Feasibility — How practical is AI for this process? How much time, effort and expense would it take?

Prioritize AI application to IAM capabilities primarily on value but gated by feasibility.

Data — The Fuel for Intelligence



What About Autonomous AI?

Full automation and maximum speed using autonomous AI agents is a valid and viable methodology.

However:

- **Never step 1**
Only enable for processes for which AI has been proven to meet all requirements.
Start with human in the loop.
- **Always tightly scoped**
Only enable incrementally for scopes for which AI decisions are high quality.





Maximize the value of your AI adoption for IAM by targeting only fit-for-purpose use cases for which you have sufficient data.

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Where Is Agentic AI Not Fit for Purpose?

Criteria	Good for AI?
Can be done well (enough) by humans: How to do it is generally known	
Priority is acceleration	
Requires clear human accountability	
Priority is quality	
Requires real-time or near-real-time responsiveness	
Requires deterministic decisions	

Where Is Agentic AI Not Fit for Purpose?

Making runtime access control decisions

Authentication

- Requires high performance
- Must be a deterministic decision

Authorization

- Requires high performance
- Must be a deterministic decision

Agentic AI could be used to improve access policies or provide insights used by access control, but NOT to make runtime access control decisions.

Low Average Value AI Use Cases

Use case	Value	Evidence
Access analysis for approval recommendations	Low	Doesn't truly automate access — makes a painful process incrementally less painful
Automated schema mapping between IGA and integrated apps	Low	Generally infrequent task (not high volume) in overall IAM workload, so value of automation is not high
Automated configuration generation for IAM system administration deployment, update and operations tasks	Low	Again, infrequent task (not high volume), so total value of this automation is not high in most organizations



To optimize IAM team AI adoption efforts, avoid using AI for runtime access control decisions and postpone investment in lower value use cases.

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High Value AI for IAM Use Cases

Use case	Feasibility
Access modeling for authorization rules and least privilege	Moderate
ATO mitigation	Moderate
Posture issue identification & analysis	High
Report generation	High

Access Modeling



Definition:

Analyze access patterns for recommendation and potential automated configuration of authorization rules for use at either admin time or run time.



Why high value:

Substantially reduces manual work for access approval and certification and accelerates automation for IGA and runtime policy. It also improves least-privilege violation detection.



Key considerations for implementation:

Effectiveness will be target-system-specific and highly dependent on identity data availability and quality. Plan for necessary data improvements and start with higher volume target systems.

Account Takeover (ATO) Mitigation



Definition:

Application of AI to improve account takeover (ATO) prevention and related adaptive access effectiveness including improving risk and threat signals.



Why high value:

Authentication compromise remains the primary and highest risk threat vector for most organizations. Any and all improvements to ATO mitigation provide substantial risk reduction benefit.



Key considerations for implementation:

Overall effectiveness depends on identification of highest value risk and threat signals and integration of these signals to the adaptive access process.

Posture Issue Identification and Analysis



Definition:

Identification and severity/impact analysis for IAM system and control configuration issues for all IAM systems and processes (except access permissions included in access modeling above).



Why high value:

Enables more effective and rapid identification of inherent risk in current IAM control posture which further enables effective action to mitigate these risks.



Key considerations for implementation:

Effectiveness will be IAM technology/tool specific as this requires the AI to learn how configuration is handled in each IAM system along with differentiating strong configuration from weak.

Report Generation



Definition:

Use of AI with natural language input to produce required reports for any reporting use case including, but not limited to, audit response and IAM program effectiveness.



Why high value:

Most IAM teams have high demand for reporting and report generation, and reporting configuration and capabilities vary across IAM tools.



Key considerations for implementation:

Effectiveness will be IAM system specific as this requires the AI to interact with the reporting functionality of each involved system.

High Value ML for IAM Use Cases

Use case	Feasibility
Identity verification	High
Biometric authentication	High
Behavior analytics	Moderate
Peer group analysis	Moderate

- Typically built-in vs. an add-on option.
- Use quality of ML implementation as a selection criteria for the broader solution.



Deliver AI adoption and investment value within your IAM program by starting with known high value use cases.

Recommendations

- ① Always evaluate AI for fit for use before even piloting it for any given IAM use case.
- ① Include planning for data management improvements along with your AI implementation planning to maximize likelihood of success.
- ① Prioritize AI for IAM implementation opportunities by value and then gated by feasibility to focus on high value use case.
- ① Do not attempt AI for runtime access control decision making. This is not fit for purpose for AI.
- ① Evaluate Gartner's recommended high value AI for IAM use cases first.

Recommended Gartner Research

To learn more about access to Gartner research, expert analyst insight, and peer communities, contact your Gartner representative or click on “Become A Client” on gartner.com to speak with one of our specialists.

🔍 [Tool: AI Use-Case Assessment for Executives](#)

Whit Andrews

🔍 [4 Data Management Practices to Improve IAM Capabilities](#)

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